Sahaj Bhadja

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Summary

AI-native full-stack engineer with 2+ years building production applications using Python, FastAPI, React, TypeScript, PostgreSQL, and cloud platforms. Experienced in LLM orchestration, human-in-the-loop workflows, state machines, RBAC, real-time updates, testing, monitoring, and graceful failure recovery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs and Azure SQL services for 500+ users, designing indexed schemas and secure workflows while reducing production response latency from 1.2 seconds to 350 milliseconds.
- Restructured a fault-tolerant LLM orchestration system with explicit plan states, approval checkpoints, failure isolation, and recovery, processing 1,000+ AI jobs while tripling throughput and improving observability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, implementing secure document ingestion, authentication, human-reviewed AI grading workflows, and error handling while resolving production issues affecting more than 150 users.
- Built React progress workflows and multithreaded processing for 1,000+ files with relational persistence, automated tests, monitoring, and failure recovery, reducing turnaround 60% under concurrent production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python and PyTorch pipelines processing 100,000+ trajectory records, improving planning accuracy 18% through reproducible experiments, model evaluation, automated validation, performance monitoring, and systematic technical debugging.
- Developed modular context-processing components combining map geometry and agent state, improving closed-loop stability 15% while documenting software interfaces and carefully validating outputs across diverse simulated failure scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable Python services and SQL integrations using AWS Glue, S3, filtering, pagination, and caching, improving response times 30% and processing efficiency 68% across multi-tenant enterprise applications.
- Maintained 20+ unit and integration suites with Jenkins, PyTest, and Mockito, accelerating bug verification, code reviews, failure-mode testing, and reliable CI/CD releases across distributed cloud environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed TypeScript and Node.js services across eight microservices with authentication, RBAC, REST APIs, WebSocket updates, and state synchronization, securely processing more than two million Azure SQL records.
- Automated functional tests, staged deployments, and service monitoring through Jenkins, Docker, and Azure, reducing release time 50% while improving graceful failure recovery, deployment consistency, and overall production availability.

Skills

Languages & Frontend: Python, TypeScript, JavaScript, SQL, React, Next.js, Node.js, HTML, CSS, component workflows, state management, forms, tables, multi-step interfaces
AI & Backend: LLM APIs, agentic workflows, human-in-the-loop approvals, FastAPI, Flask, REST APIs, WebSockets, state machines, PostgreSQL, Azure SQL, SQLAlchemy familiarity, schema design
Identity & Quality: Authentication, RBAC, access controls, secure APIs, unit testing, integration testing, PyTest, Mockito, error handling, failure isolation, graceful degradation, monitoring, audit logging
Cloud & Delivery: AWS, Azure, GCP, S3, Git, branching, pull requests, code reviews, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes, Agile, technical documentation

Projects

Big-Brain | Live | *Python, React, PostgreSQL, LLMs, FAISS, GCP*
- Built an AI document platform over 50,000+ files using React, Python APIs, PostgreSQL, transformer retrieval, approval-oriented validation, optimized indexing, and cloud deployment with sub-second contextual search.

PiXMod | Live | *React, Next.js, JavaScript, MongoDB, Stripe*
- Built a responsive full-stack product using React, Next.js, MongoDB, and Stripe, implementing maintainable user workflows, API integrations, automated validation, and subscription features that increased premium conversion 18%.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating secure REST APIs, real-time visualizations, MongoDB queries, Kubernetes deployment, production monitoring, and automated release workflows.